



Product Requirements Document: AI-Powered Agricultural Data Loader

The Problem

PepsiCo agricultural growers (managing up to 300 fields per season) and internal Agronomists face a massive data collection bottleneck. The core issue is a manual, frustrating translation process:

1. Speaking Different Languages (The Translation Problem): Growers export data from their tractors (e.g., John Deere) which uses terms like "Sowing Date" or "DKC61-69". PepsiCo's database requires specific terms like "Date of Planting" or "Variety Master". Today, a human must manually read the John Deere report and type the translated data into PepsiCo's system.

- *Current Time Impact:* It takes roughly 15-20 minutes to manually cross-reference and type the data for a single field. For a 300-field farm, this is 75 to 100 hours of manual data entry per season.

2. Too Many Systems to Manage: Growers globally use between 10 and 15 different farming tools (MyJohnDeere, Climate FieldView, Cropwise). PepsiCo cannot build custom software integrations for all 15 tools, leaving manual data entry as the only current fallback.

3. Delayed, Inaccurate Data: Because this data entry is so painful, growers put it off until the absolute deadline at the end of the season.

- *Current Accuracy Impact:* Manual typing leads to "fat-finger" typos (e.g., typing 32,000 instead of 34,000 for a seeding rate). This requires agronomists to spend weeks emailing growers to fix errors, with initial accuracy often dipping below 80%. This delay starves PepsiCo's sustainability tracking and predictive yield models.

Vision & Impact

Our vision is to build an intelligent, AI-powered "Smart Translator" that turns a grueling 100-hour data entry chore into a 15-minute review process.

Expected Impact:

- **Drag-and-Drop Simple:** Growers simply upload their raw reports (PDFs, Excel spreadsheets) into our app. The AI acts like a human assistant, reading the document and automatically filling out PepsiCo's digital forms.
- **Massive Time Savings:** We shift the grower's job from *Data Entry* to *Data Approval*.
 - *Future Time Impact:* What used to take 100 hours of typing will take 15 minutes of reviewing the AI's work and clicking "Approve."
- **Guaranteed Data Accuracy:** The AI checks for typos and outlier numbers *before* the data is submitted.
 - *Future Accuracy Impact:* Data accuracy jumps from 80% to 100%, eliminating the endless back-and-forth emails between Agronomists and Growers.

Personas: Who are we building for?

To truly understand the problem, we have to look through the lens of the people actually sitting in the farm office.

1. The Farm Operations Manager / Data Admin (The Primary User)

- **"Who I am":** "I run the back office for our family farm. We manage over 300 fields. Our operators are out in the cabs driving John Deere tractors, and everything they do is tracked perfectly in our John Deere software."
- **"My Reality (The Pain)":** "When a season ends, John Deere gives me a beautiful PDF summary. But PepsiCo's system doesn't talk to John Deere. So, I have to split-screen hundreds of these PDFs, log into the PepsiCo portal, and manually type those numbers in over and over. It takes me weeks of mind-numbing data entry."
- **"What I Need":** "I already collected this data. Don't make me do it twice. Let me just drag and drop my John Deere PDFs, have your AI figure out what PepsiCo wants, and let me just click 'Approve'. I want to be done in 15 minutes."

2. The Farm Owner / Head Grower

- **"Who I am":** "I own the operation. I care about crop yield, weather, and market prices. I don't want to be sitting behind a desk."
- **"My Reality (The Pain)":** "We work with dozens of buyers. Every single one of them has a different portal and a different template. It's a massive distraction from farming, and I have to pay for extra administrative help just to stay compliant."
- **"What I Need":** "If PepsiCo makes it effortless to report my data, they become my preferred, easiest partner to work with."

3. The PepsiCo Agronomist (The "Receiver")

- **Profile:** Responsible for regional crop yield and sustainability compliance.
- **Pain Point:** Spends hours "chasing data" from frustrated farm managers. Spends even more time finding bad data (like incorrect chemical application rates) and emailing the farm to fix it.
- **Motivation:** Needs clean, error-free data instantly to generate accurate yield forecasts and sustainability reports without annoying the growers.

Critical User Journeys (CUJs)

CUJ 1: End-of-Season Harvest Reporting (The "Bulk Load")

- **Trigger:** The Farm Admin finishes the fall harvest and exports a standard "Overview Harvest" PDF from their John Deere system.
- **Action:** The Admin drags the PDF into the PepsiCo AI App.
- **System Response (The "Maker"):** The AI reads the document. It finds "Yield: 104 bu/ac" and translates it to PepsiCo's format. It flags anything it is unsure about (like a seed variety name) in **Amber ("Needs Review")**.
- **Resolution (The "Checker"):** The Admin views the screen, selects the correct PepsiCo variety from a dropdown to fix the amber flag, and clicks "Submit."
- **Time Saved:** 20 minutes of typing reduced to 15 seconds of clicking.

CUJ 2: Mid-Season Chemical Application (Catching Errors Early)

- **Trigger:** The Farm Admin uploads a "Tank Mix Rate" PDF to report a recent pesticide spray.
- **System Response (The "Maker"):** The AI reads that "9.95 gallons/acre" were applied. The AI knows PepsiCo needs this in metric units (Liters/Hectare) and does the math automatically. It also checks PepsiCo's rules and sees this amount is normal, highlighting it in **Green ("Looks Good")**.
- **Resolution (The "Checker"):** The Admin trusts the Green highlight, clicks "Approve", and immediately fulfills their reporting requirement with 100% accuracy.

UI & UX Recommendations (By Persona)

To guarantee adoption, the interface must look and feel completely different for each persona based on their specific goals.

1. Farm Operations Manager (The Desktop "Reviewer" UI)

- **Design Philosophy:** "Traffic Lights & Spreadsheets." This user is comfortable with data but hates typing. The interface should feel like a smart, pre-filled inbox.
- **The UX Flow:** 1. A massive "Drag and Drop Files Here" zone in the center of the screen.
2. A split-screen view: the original John Deere PDF is visible on the left for easy reading, and the PepsiCo digital form is on the right.
3. **Traffic Light Color Coding:** Fields auto-filled by AI are highlighted in **Green** (100% matched to PepsiCo's dictionary). Fields the AI is unsure about are highlighted in **Amber**.
- **The Benefit:** Instead of staring at a blank form trying to remember what to type, the user's eye is immediately drawn only to the Amber fields. They click a dropdown, fix the discrepancy, and hit "Approve All."
- **Impact Visualized:** The screen clearly shows: *"AI saved you 14 hours of data entry today."*

2. Farm Owner / Head Grower (The Mobile "Peace of Mind" UI)

- **Design Philosophy:** "Green Checks and High-Level Summaries." This user wants to know they are compliant and get back to work.
- **The UX Flow:**
 1. A mobile-friendly, ultra-simple dashboard.
 2. A prominent status ring: "Your farm is 100% compliant with PepsiCo reporting for the 2026 Season."
 3. A quick summary card: *"Your team uploaded 15 harvest records this week. 0 errors detected."*
- **The Benefit:** Zero data-entry interface. They don't see rows and columns; they see a trusted partnership tool that tells them their business is running smoothly.

3. PepsiCo Agronomist (The "Command Center" UI)

- **Design Philosophy:** "Exception Management." Agronomists shouldn't look at data unless something is wrong.
- **The UX Flow:**
 1. A map or list view of all assigned farms, sorted by reporting progress.
 2. A "Data Health" score for each farm.
 3. An **Exceptions Inbox:** Instead of downloading 50 spreadsheets to find errors, the Agronomist only sees records that failed the AI's math rules (e.g., a grower reported a yield of 900 bu/ac, which is biologically impossible).

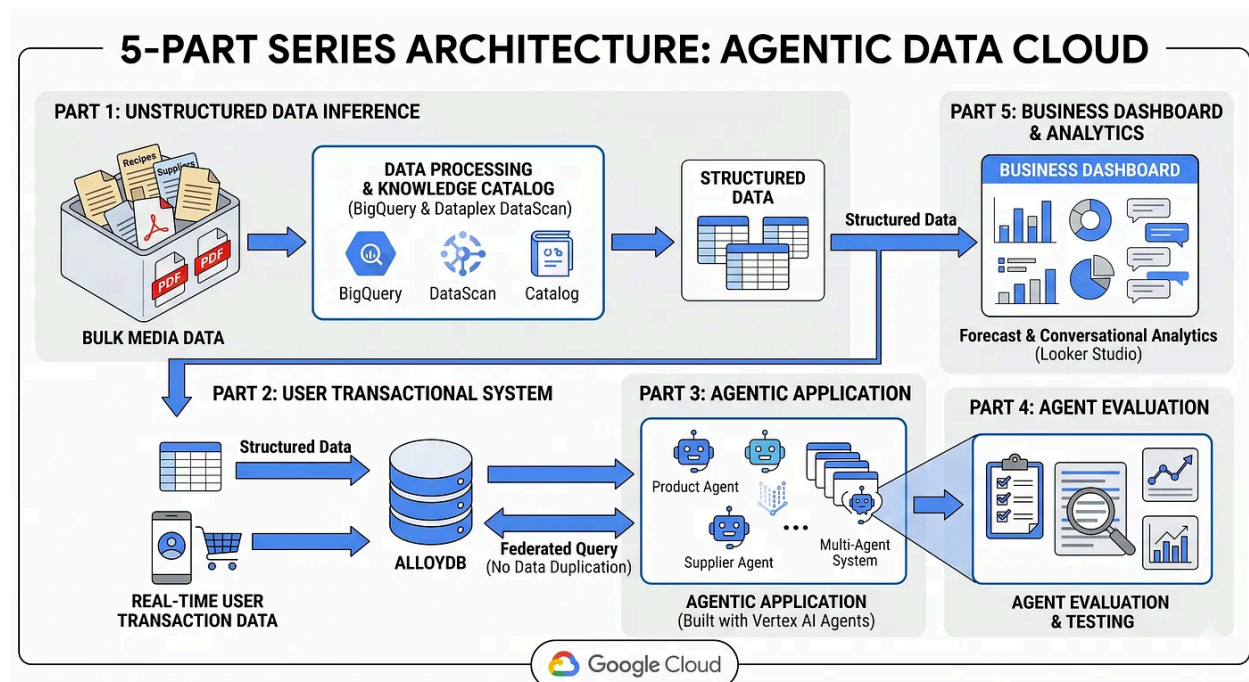
- **The Benefit:** Eliminates the need to manually audit every row of data. The Agronomist only intervenes when the AI flags a true anomaly.

The Solution (How It Works Behind the Scenes)

Instead of building 100 different custom software connections, we are using Google Cloud AI as a universal translator.

Prototype Architecture Flow (User Perspective)

The following diagram illustrates the simple, left-to-right journey of a grower's data from their local computer, through our AI translator, and into PepsiCo's system.



graph LR

%% Define Nodes

A[Farm Admin / Grower]

B[1. Document Upload
 Drag & Drop App]

C{2. The Smart Translator
 Google Gemini AI}

D[3. The Review Screen
 Maker-Checker UI]

E[(4. Final Storage
 PepsiCo Database)]

%% Define Connections

A -->|Drops John Deere PDF| B
B -->|Reads Text/Data| C
C -->|Translates to PepsiCo Format
 Converts Units & Checks Math| D
D -->|User Fixes 'Amber' Flags
 & Clicks Approve| E

%% Styling

```
classDef user fill:#e1f5fe,stroke:#01579b,stroke-width:2px,color:#000;  
classDef step fill:#f5f5f5,stroke:#616161,stroke-width:2px,color:#000;  
classDef ai fill:#e8f5e9,stroke:#2e7d32,stroke-width:2px,color:#000;  
classDef ui fill:#fff3e0,stroke:#e65100,stroke-width:2px,color:#000;  
classDef db fill:#f3e5f5,stroke:#4a148c,stroke-width:2px,color:#000;
```

```
class A user;  
class B step;  
class C ai;  
class D ui;  
class E db;
```

The Technology Flow (Plain English)

1. **The Document Reader (Google Cloud Storage & Dataplex):** The grower uploads their PDF or Excel file. Google Cloud securely stores it and prepares it to be read.
2. **The Smart Translator (Gemini AI in BigQuery):** We give the Gemini AI the PepsiCo "Master Dictionary." The AI reads the grower's PDF, understands that "Start Date" means "Date of Planting," and automatically organizes the scattered text into neat, structured rows of data.
3. **The Review Screen (Application Integration):** Before anything is saved permanently, the organized data is sent to a simple web page. The grower sees exactly what the AI found and must "Check the box" to approve it.
4. **Final Storage (Google Cloud BigQuery):** Once approved by the human, the 100% accurate data is sent securely to PepsiCo's main data warehouse (the Enterprise Data Flow natively hosted on Google Cloud BigQuery).

Phasing & Milestones

- **Phase 1 (Proof of Concept):** Build a prototype that proves the AI can accurately read the 3 most common John Deere PDFs (Seeding, Application, Harvest) and match them to PepsiCo's core requirements.
- **Phase 2 (Pilot):** Give the simple Review App to a small group of "Lead Growers" to test how much time it saves them in the real world.
- **Phase 3 (Global Rollout):** Connect the app directly to PepsiCo's centralized Google Cloud BigQuery database for global use.

Success Metrics

- **Time Saved for Growers:** Reduce total seasonal reporting time from 100 hours to under 1 hour.
- **Data Velocity:** Reduce the time it takes PepsiCo to get data after a field event from 60 days to under 1 day.
- **Accuracy Improvement:** Reduce agronomist data rejection rates by 80% by catching errors and standardizing units before submission.

Outstanding Questions & Next Steps

Strategic Questions for Prototype Discovery

- **Verification Source of Truth:** Should the Review Screen show the original PDF side-by-side with the AI's extracted data so the grower can double-check the numbers easily?
- **Accuracy Detection Today:** How does PepsiCo currently detect inaccuracies? Are there specific math rules (like maximum yield limits) we should program the AI to check against?
- **Data Utilization:** Once the data is verified, what is the immediate next step? Do we need to generate a PDF receipt for the grower to keep for their own compliance records?